



POLICY BRIEF

Indigenous Natural Resources Management among Pastoral Communities: Old Solutions for New Challenges

Oliver Vivian Wasonga¹, Jacob Olonde¹, Staline Kibet¹

¹Department of Land Resource Management and Agricultural Technology, University of Nairobi
P.O. Box 29053-00625, Nairobi

²Christensen Fund, African Rift Valley

Summary

Pastoralism employs diverse indigenous knowledge and practices in management of natural resources, and is known to be the most viable production system in the precarious arid and semi-arid environments. Most pastoral land in Kenya is communally owned with traditional institutions and structures that regulate access to

grazing and water resources. This brief examines the role of indigenous knowledge on natural resource management; reviews traditional institutions that govern resource use and factors underpinning the indigenous knowledge and its use in management of natural resources.

Pastoralism has evolved in Northern Kenya

Pastoralism has evolved in the uncertain and ever-changing environments and socio-economic conditions through application of indigenous knowledge in management of natural resources and meeting their social, cultural and economic obligations.

In the case of Northern Kenya, traditional NRM entails use of customary institutions that regulate access to resources through zoning of grazing land into wet, dry and drought grazing areas and setting rules to govern use to ensure sustainability.

The customary institutions enforce such strategies as range zoning which help maintain good health of the environment through rotational use of grazing resources to allow adequate rest before subsequent use.

Communal ownership of resources and indigenous ecological knowledge are critical components that enable pastoralism to thrive in the harsh and uncertain environments. Communal land tenure enables pastoralists to employ mobility to track and optimally use pastures and water that are unevenly distributed in space and time. Pastoralists possess



extensive knowledge of livestock mobility, improvement of range condition through burning; pasture reservation, herd splitting and regulation of utilization of woody species.

The brief presents the pastoral indigenous ecological knowledge on natural resource management

Why Indigenous NRM?

- Has a zoning element which allows use of the transient grazing and water resources at their peak of production.
- Promotes communal land tenure: enables mobility knowing that resources asymmetrically distributes in space and time
- Enhances social networks: aids herd mobility through reciprocal access to pastures beyond a community's boundaries
- Encourages sound ecological knowledge: guides on decisions of when and where to move; allows assessment of range condition through

Rangeland classification forms the basis for traditional natural resources management

In pastoral areas, traditional natural resource management is based on sound understanding of the pastoral landscapes and their potential and suitability for livestock use. Rangeland classification by pastoralists forms the basis for use and management. The classification is based on several bio-



and the implications of the socioeconomic, political and climatic changes for their application in contemporary world. Pokot pastoralists of Baringo and East Pokot Counties

Partitioning of grazing land by the Borana Community of northern Kenya

The Borana community categorizes the grazing landscapes into three main landscapes, such as the laf Bona, gamoji and (laf oolat/goli oola): using a combination of climate (rainfall and temperature), soil, topography, and vegetation.

Dry season grazing area (laf bona): Characterised by permanent water points — mainly rivers, springs, shallow wells, water pans and boreholes — the plant species here are long-lived and evergreen.

Wet season grazing areas (gamoji): Characterised by abundant but transient sub-surface water resources, these are normally found in low-lying areas, which are warm in the rainy seasons and have abundant grass compared to trees. These areas are only accessed every five to six months, so twice a year — between the end of March and end of May, and from October to December.

Drought grazing reserves (laf oolat/goli oola): This refuge area is only accessed during extreme pasture scarcity, when the rains fail and the dry season grazing areas have been exhausted. The drought reserves are dominated by perennials (mar idha), which are capable of supporting livestock for a long time.

Source: (Wasonga et al. 2016)

physical factors that include climate, topography, soil characteristics, vegetation type and common plant species in specific areas. Sites may be assigned a name based on common plant species, soil type or color, topography, name of area close to the grazing site and historic events.



Grazing in the pastoral landscapes is influenced by a number of factors. Some of these, listed in the box below, are human determined such as the



number of animals and ethnic relations, while others are dictated by natural circumstances such as droughts and diseases.

Factors influencing Grazing

- Macro landscapes with microclimates and seasonal patterns,
- Suitability for livestock use, grazing potential and botanical composition,
- Soil properties and area's susceptibility to degradation,
- Security: e.g. security and conflicts; relations with neighboring communities;
- Availability of herding labour;
- Existence of customary institutions that regulate access to natural resources

Livestock Mobility is important in tracking variable water and pasture

Livestock mobility is a strategy used by pastoralists to track range resources particularly water and pastures that are spatially and temporally variable. This strategy relies on a wealth of tacit knowledge accumulated over generations and passed on through apprenticeship where young persons accompanied older herders during grazing cycles or as part of whole family practice nomadic/transhumance lifestyle learn by doing.

Apart from observable livestock attributes, pastoralists use livestock instinct to decide when to move their animals. In many instances, livestock reluctance to return home early, tendency to graze at night and unrest during milking reflect poor condition of the grazing land.

Observable Livestock Attributes Influencing Mobility

- Rumen-fill, milk yield, texture of animal dung, coat condition, weight gain, animal health, mating frequency and successful calving or kidding rates.
- Livestock grazed on good pasture are expected to show good health, high mating frequency, rapid weight gain, successful calving rate, increase in milk yield, smoother fur and deep breathing at night.
- The reverse is true for poor range condition as decline in these attributes will trigger migration to a new site.

Source: Dabasso et al 2012)



Herd Management ensures spatial distribution of livestock within the larger landscape

Herd management in pastoral systems involves livestock division of household herds to minimize losses during drastic environmental changes, raids and diseases. Apart from being a risk-minimizing strategy, herd splitting ensures spatial distribution of livestock within the larger landscape to ensure even use of grazing and water resources.

The Rendille, for example, has two types of herd systems, namely *goob* and *fora*. *Goob* system is the

camp or home-based herd, while *fora* represent the satellite herd (Refer to Roba and Oba 2009). When dry season sets in, the camp remains with a small herd consisting of milking cows, young and sick animals, while *fora* consists of the larger dry herd. When the rain falls, there is plenty of pasture and water resources that can maintain both the mobile and home-based herds, and therefore herders with satellite herds return to the village.

Pastoralists have elaborate knowledge systems of burning to control their ecosystem

Traditionally, pastoralists used fire as a way of managing their ecosystem. Basically, controlled burning was used to improve the growth of desirable pastures, reduce woody growth and kill pests such tick and fleas. The Borana of southern Ethiopia used burning to improve pasture nutritional quality and accessibility of fodder. The general increase in bush encroachment in areas of northern Kenya can be attributed to the ban on use of fire and this has affected the ratio of livestock species kept by pastoral communities. There is significant

increase in the number of goats with respect to cattle in the northern Kenya due to the increased abundance of browser plant species as well as climate change.

Pastoralists have elaborate knowledge systems of burning to control their ecosystem to desirable conditions. For instance, the Maasai burn their pasture during the evening in the absence of strong winds to prevent uncontrollable fires from occurring.

Customary governance institutions are responsible for regulating access to and the use of natural resources

Customary institutions have been defined as laws and norms, both formal and informal that facilitates social interactions by allowing people to co-

operate and achieve common objectives for the common good. One such customary institution is



the Borana council of elders, or jars deedha among Borana of Southern Ethiopia and Northern Kenya.

Made up of opinion and religious leaders selected by the community, this customary governance institution is responsible for regulating access to and the use of natural resources to ensure that the rangelands are used sustainably and provide multiple benefits



Members are guided by regulations set and respected by the community: customary laws derived from the gada, the supreme Borana governance structure that preserves traditional laws and codes of conduct with amendments and additions based on the evolving environmental, social and cultural context (refer to Wasonga et al 2016).

Roles and Responsibilities of Jars Deedha

- Ensures the sustainable use of rangelands, by striking a delicate balance between livestock numbers, water supply and the amount and quality of standing pasture within range of the water points.
- Defines grazing areas and controls water and pasture access.
- Ensures that every household separates their herd into two — home-based herds of milking, sick and young stock and the **fora** (satellite camp) herds, which are strong and healthy — and reinforces the grazing pattern.
- Restrict grazing in these areas when alternative grazing resources are available, ensuring sustainable resource use to help make the pastoral environment and livelihoods more resilient.

Source: Wasonga et, al, 2016

Traditional system has been instrumental in determining the lifeline of pastoral livelihoods

The use of indigenous natural resources management in pastoralism has endured numerous challenges. For decades, the traditional system has been instrumental in determining the lifeline of pastoral livelihoods, survival and resilience to ca-

lamities. However, with the changing political governance, increased urbanization and socialization, as well as the increased exposure to climate change related disasters, the system has faced a lot of discomfort.



Policy Implications/ Recommendations

As discussed in the policy brief, in order to cement the traditional natural resources management system among the pastoral communities, the following recommendations will go a long way in offering the much needed interventions:

- Establish and strengthen inter-community and cross-county grazing committees to address the influx of livestock from neighbouring areas that put pressure on the resource base.
- Addressing climate change impacts by introducing climate resilient fodder and livestock varieties
- Creating awareness and sensitizing communities on available options and facilitate the shift if necessary
- Invest in mass water storage facilities such as dams with specific watering points for both dry and wet seasons.
- Control upstream water abstraction to allow free water flow downstream

Reference

Barrow E and Mlenge W. (2003). Trees as key to pastoralist risk management in semi-arid landscapes in Shinyanga, Tanzania and Turkana, Kenya. Paper presented at The International Conference on Rural Livelihoods, Forests and Biodiversity 19-23 May 2003, Bonn, Germany.

Butz, R. J. (2009). Traditional fire management: historical fire regimes and land use change in pastoral East Africa. *International Journal of Wildland Fire*, 18(4), 442-450.

Fratkin, E. (2005). A Comparative Approach to Transition and Social Change among Livestock Pastoralists in East Africa and Central Asia

Roba, G. M. (2013). *Indigenous Institutions and Knowledge in Natural Resource Management in Project Pilot Districts of Mbeere-north, Kyuso, Narok-north and Dadaab: Final Report on Documentation of Indigenous Institutions*. United Nations Development Programme.

Vehrs, H. P. (2016). Changes in landscape vegetation, forage plant composition and herding structure in the pastoralist livelihoods of East Pokot, Kenya. *Journal of Eastern African Studies*, 10(1), 88-110.

Wasonga, V. O., Ngugi, R. K., and Kitalyi, A. (2003). Traditional range condition and trend assessment: lessons from Pokot and Il Chamus pastoralists of Kenya. *Anthropologist*, 5(2), 79-88.